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Characterization of stable isotope fractionation during methane production in the sediment of a eutrophic lake, Lake Dagow, Germany

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Abstract

We measured $\delta^{13}C$ of CO_2, CH_4, and acetate-methyl in profundal sediment of eutrophic Lake Dagow by incubation experiments in the presence and absence of methanogenic inhibitors chloroform, bromoethane sulfonate (BES), and methyl fluoride, which have different specificities. Methyl fluoride predominantly inhibits acetoclastic methanogenesis and affects hydrogenotrophic methanogenesis relatively little. Optimization of methyl fluoride concentrations resulted in complete inhibition of acetoclastic methanogenesis. Methane was then exclusively produced by hydrogenotrophic methanogenesis and thus allowed determination of the fractionation factors specific for this methanogenic pathway. Acetate, which was then no longer consumed, accumulated and allowed determination of the isotopic signatures of the fermentatively produced acetate. BES and chloroform also inhibited CH_4 production and resulted in accumulation of acetate. The fractionation factor for hydrogenotrophic methanogenesis exhibited variability, e.g., it changed with sediment depth. The $\delta^{13}C$ of the methyl group of the accumulated acetate was similar to the $\delta^{13}C$ of sedimentary organic carbon, while that of the carboxyl group was by about 12‰ higher. However, the $\delta^{13}C$ of the acetate was by about 5‰ lower in samples with uninhibited compared with inhibited acetoclastic methanogenesis, indicating unusual isotopic fractionation. The isotope data were used for calculation of the relative contribution of hydrogenotrophic vs. acetoclastic methanogenesis to total CH_4 production. Contribution of hydrogenotrophic methanogenesis increased with sediment depth from about 35% to 60%, indicating that organic matter was only partially oxidized in deeper sediment layers.

Eutrophic lakes usually develop an anoxic hypolimnion that lasts over much of the summer season. During the rest of the year, when the hypolimnion is oxic, only the top few centimeters of the sediment become oxidized while the rest remains reduced. In these sediments, production of CH_4 is the terminal step in the degradation of organic matter, and carbon cycling in the lakes is greatly affected by the release of CH_4 from the sediment into the water column (Rudd and Taylor 1980). Organic matter is fermented to acetate, H_2, and CO_2, which are subsequently converted to CH_4. The relative contribution of acetate and H_2 plus CO_2 to CH_4 production can vary, and a mechanistic explanation for this behavior is not always available (Conrad 1999). The relative contribution of either acetoclastic or hydrogenotrophic methanogenesis also affects the $\delta^{13}C$ of the produced CH_4 (Whiticar 1999; Hornibrook et al. 2000). During hydrogenotrophic methanogenesis ($4H_2 + CO_2 \rightarrow CH_4 + 2H_2O$) the isotopically lighter carbon is strongly preferred, whereas the isotope effect is less expressed in acetoclastic methanogenesis ($CH_3COOH \rightarrow CH_4 + CO_2$), in which CH_4 is almost exclusively produced from the methyl group of the acetate (Weimer and Zeikus 1978; DeGraaf et al. 1996). Hence, the wide range of different $\delta^{13}C$ of CH_4 that are found in various wetlands and other anoxic environments may mostly be due to the methanogenic pathways in operation. This in turn greatly affects the computation of global methane budgets using isotopic data of CH_4 sources as constraint (Fung et al. 1991; Quay et al. 1991; Tyler 1992). Although there is already an extensive literature on $\delta^{13}C$ of CH_4 and CO_2 in natural environments (Whiticar et al. 1986; Breas et al. 2001; Conrad 2005), these data are largely descriptive and do not provide a basic understanding of the underlying microbial processes. To gain more knowledge we decided to study the paths of CH_4 production and the isotopic fractionation of $^{13}C:^{12}C$ using sediment of a eutrophic lake.

The difference in isotopic fractionation during hydrogenotrophic and acetoclastic methanogenesis can in principle be used to calculate the relative contribution of the two methanogenic pathways to total CH_4 production, when the following stable isotopic signatures and isotopic fractionation factors are known for the environmental system (Conrad 2005): $\delta^{13}C$ of CH_4 (δ_{CH_4}), CO_2 (δ_{CO_2}), and acetate-methyl ($\delta_{ac\text{-}methyl}$), as well as the fractionation factors (α) (or equivalently the enrichment factors ε) for the reduction of CO_2 to CH_4 (α_{CO_2/CH_4}; ε_{CO_2/CH_4}) and of the acetate-methyl to CH_4 ($\alpha_{ac\text{-}methyl/CH_4}$; $\varepsilon_{ac\text{-}methyl/CH_4}$). However, fractionation factors can vary considerably from one environment to the other, for example, because of different

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microbial communities (Chidthaisong et al. 2002; Penning et al. 2006*a*) or different energetic conditions for methanogenesis (Penning et al. 2005; Valentine et al. 2005). Thus, values of α_{CO_2/CH_4} can vary between 1.03 and 1.09 (equivalent to ε_{CO_2/CH_4} of −90‰ to −30‰). Experimental determination of fractionation factors in environmental samples is difficult, since either the hydrogenotrophic or the acetoclastic methanogenic pathway must be suppressed in order to determine the isotope fractionation by one of the two pathways specifically (Conrad 2005). Methyl fluoride (CH_3F) was reported to be a specific inhibitor of acetoclastic methanogens, while hydrogenotrophic methanogens were not affected (Janssen and Frenzel 1997). This observation was recently confirmed for the complex methanogenic communities living on rice roots (Penning et al. 2006*b*) and in rice field soil (Penning and Conrad 2007), thus allowing the determination of δ_{CH_4} specifically produced from CO_2 reduction (δ_{mc}) and of α_{CO_2/CH_4} (or ε_{CO_2/CH_4}). The specific inhibition of acetoclastic methanogenesis furthermore allowed the determination of the $\delta^{13}C$ of the fermentatively produced acetate (δ_{ac}) and acetate-methyl ($\delta_{ac\text{-}methyl}$) (Penning et al. 2006*b*; Penning and Conrad 2007).

The fractionation of $^{13}C:^{12}C$ during CH_4 production from acetate (α_{ac/CH_4}) is smaller than that from H_2 plus CO_2, typically on the order of 1.01 to 1.03 (equivalent to ε_{ac/CH_4} of −30‰ to −10‰) (Gelwicks et al. 1994; Penning et al. 2006*a*). Frequently, there is no fractionation at all, since acetate concentrations in lake sediments are usually very low (<5 μmol L^{-1}), and if each molecule of acetate that is produced is subsequently consumed, fractionation is zero ($\varepsilon_{ac/CH_4} = 0\%$; $\alpha_{ac/CH_4} = 1$).

Here, we comparatively studied $^{13}C:^{12}C$ isotopic fractionation during CH_4 production in the presence and absence of inhibitors. The objectives of our study were to learn more about the processes involved in fractionation of stable carbon isotopes during the complex degradation of organic matter to CH_4 and to apply the isotopic data to quantify the relative role of acetoclastic and hydrogenotrophic methanogenesis in eutrophic Lake Dagow sediment. Our data revealed a nonnormal isotopic fractionation of acetate and showed that hydrogenotrophic methanogenesis was the predominant CH_4 production pathway.

Methods

Sampling procedure—Eutrophic Lake Dagow is located in northern Brandenburg, Germany. The main morphological and limnological characteristics of the lake were described by Casper (1996). The temperature and dissolved oxygen in the water column (9-m deep) were measured with an Oxi 197 probe (WTW) immediately before taking sediment samples. Sediment was sampled with an Eckman dredge on 04 Jul 2001, 17 Aug 2005, and 04 Jul 2006, and the top layer (0–5 cm) transferred into a Nalgene™ plastic bottle, which was completely filled and stored at 4°C until usage, which was after periods of 8 months, 1 week, and 4 months, respectively. Six sediment cores (diameter 7.5 cm) were sampled using a Jenkin-sediment-sampler (Ohnstad and Jones 1982) each on 05 May and 08 Aug 2003. The cores were sliced and processed within 1 h. The sediment cores were sliced into subcores (0–3, 3–6, 6–10, 10–15, 15–20 cm) under a flow of N_2. The slices of the six cores were pooled such that three independent replicate sediment samples were obtained. Aliquots were used for incubation in the presence and absence of CH_3F. An additional core was taken on 12 Jun 2003 for determination of dry matter content, organic matter content (loss of ignition), calcium carbonate, and the $\delta^{13}C$ of sediment carbon. For the latter, ^{13}C was measured in the sediment before (δ_{tot}) and after (δ_{org}) acidification, the difference being due to carbonate (δ_{carb}) (Nüsslein et al. 2003).

Incubation experiments—For sediment incubations, 10-mL aliquots were transferred into 120-mL sterile serum bottles, flushed with N_2, and closed with butyl rubber stoppers and incubated at 10°C overnight. Then, the bottles were flushed again with N_2 and further incubated at 10°C. The gas headspace of some of the bottles was supplemented with methyl fluoride (CH_3F) (Fluorochrome company) at the desired concentration. Bromoethane sulfonate (BES) was applied as aqueous solution of the sodium salt, chloroform ($CHCl_3$) was added directly.

Gas samples were repeatedly taken from the headspace of the bottles over a period of 13–24 d and analyzed for CH_4 and CO_2 as well as $\delta^{13}C$ of CH_4 and CO_2. At the end of incubation, liquid samples were taken and analyzed for concentration and $\delta^{13}C$ of acetate. The dry weight of the sample was determined gravimetrically. Some of the sediment grab samples were set up in additional replicates, of which triplicates were opened at different time points during the incubation and the liquid analyzed as described above.

Analytical techniques—Gas samples (0.25–1.0 mL) were taken with a gas-tight pressure lock syringe (Dynatech), after the bottles were vigorously shaken by hand, and analyzed immediately by gas chromatography. CH_4, CH_3F, and CO_2 were analyzed by gas chromatography using a flame ionization detector (Shimadzu). CO_2 was detected after conversion to CH_4 with a methanizer (Ni-catalyst at 350°C, Chrompack). Liquid samples were membrane-filtered (0.2 μm) and stored frozen (−20°C) until analysis. Acetate was measured by high-pressure liquid chromatography (HPLC; Sykam) with a refraction index and ultraviolet (UV) detector, having a detection limit of about 5 μmol L^{-1}.

Stable isotope analysis of $^{13}C:^{12}C$ in gas samples was performed using a gas chromatograph combustion isotope ratio mass spectrometer (GC-C-IRMS) system (Thermoquest). Its principle operation is described by Brand (1996). The CH_4 and CO_2 in the gas samples (40–400 μL) were first separated in a Hewlett Packard 6890 gas chromatograph using a Pora Plot Q column (27.5-m length, 0.32-mm inner diameter; 10-μm film thickness; Chrompack) at 30°C with He (99.996% purity; 2.6 mL min^{-1}) as carrier gas. After conversion of CH_4 to CO_2 in the Finnigan standard GC combustion interface III, the $^{13}C:^{12}C$ was analyzed by the IRMS instrument (Finnigan MAT model Delta Plus).

The isotope reference gas was CO_2 (99.998% purity; Messer-Griessheim) calibrated with the working standard methylstearate (Merck). The latter was intercalibrated at the Max-Planck-Institut für Biogeochemie (courtesy of Dr. W.A. Brand) against NBS 22 and USGS 24, and reported in the delta notation vs. V-PDB: $\delta^{13}C = 10^3 (R_{sa}/R_{st}-1)$ with $R = {}^{13}C:{}^{12}C$ of sample (sa) and standard (st), respectively. The precision of repeated analysis was ±0.2‰, when 1.3 nmol CH_4 was injected.

Isotopic measurements of acetate were performed on a HPLC system (Spectra System P1000, Thermo Finnigan; Mistral, Spark) equipped with an ion-exclusion column (Aminex HPX-87-H, Biorad) and coupled to Finnigan LC IsoLink (Thermo Electron) as described by Krummen et al. (2004). Isotope ratios were detected on an IRMS (Finnigan MAT delta plus advantage). Isotope reference gas was CO_2 calibrated as described above.

An off-line pyrolysis was conducted to determine $\delta^{13}C$ of the methyl group of acetate ($\delta_{ac\text{-}methyl}$). Acetate in the liquid sample was purified using HPLC by collecting the acetate fraction from each run. The purified sample exhibited only one peak when analyzed by HPLC or GC. The purified sample was added to a strong NaOH solution and dried in a Pyrex tube under vacuum. The dried reactants were pyrolyzed under vacuum at 400°C, converting the carboxyl carbon to CO_2 and the methyl carbon to CH_4 (Blair et al. 1985; Penning et al. 2006*b*). Gas samples were taken and then analyzed by GC-C-IRMS as above.

The analysis of the ^{13}C in organic matter was carried out at the Institute for Soil Science and Forest Nutrition (IBW) at the University of Göttingen, Germany (courtesy of Heinz Flessa), using an elemental analyzer coupled to mass spectrometer.

Calculations—Fractionation factors for a reaction $A \rightarrow B$ are defined after Hayes (1993):

$$\alpha_{A/B} = (\delta_A + 1000)/(\delta_B + 1000) \quad (1)$$

sometimes expressed as isotopic enrichment factor $\varepsilon \equiv 10^3 (1 - \alpha)$. The isotopic signature for a newly formed CH_4 (δ_n) was calculated from the isotopic signatures at two time points $t = 1$ (δ_1) and $t = 2$ (δ_2) by the following mass balance equation:

$$\delta_2 = f_n \delta_n + (1 - f_n)\delta_1 \quad (2)$$

with f_n being the fraction of the newly formed C-compound relative to the total at $t = 2$.

The apparent fractionation factor for conversion of CO_2 to CH_4 is given by

$$\alpha_{app} = (\delta_{CO_2} + 1000)/(\delta_{CH_4} + 1000) \quad (3)$$

Relative contribution of H_2/CO_2-derived CH_4 to total CH_4 was determined using the following mass balance equation (Conrad 2005):

$$f_{H_2} = (\delta_{CH_4} - \delta_{ma})/(\delta_{mc} - \delta_{ma}) \quad (4)$$

where f_{H_2} is the fraction of CH_4 formed from H_2 plus CO_2, δ_{CH_4} the $\delta^{13}C$ of total produced methane, and δ_{ma} and δ_{mc} are the isotope ratios of CH_4 either derived from acetate or H_2 plus CO_2.

Results

Inhibition experiments with samples from the upper sediment layer—The upper sediment layer (0–5-cm depth) had the following characteristics: water content, 95–97% (w/w); carbonate, 23–39% of dry matter; organic carbon, 18–21% of dry matter; pH 7.5; $\delta^{13}C$ of organic carbon (δ_{org}), –30.1‰ ± 0.05‰; and $\delta^{13}C$ of carbonate carbon (δ_{carb}), 8.1‰ ± 0.1‰.

Experiments with sediment grab samples showed that incubation at 10°C (in situ temperature) resulted in immediate and almost linear production of CH_4. Addition of CH_3F to the headspace of the incubations resulted in partial inhibition of production of CH_4 and a decrease of the $\delta^{13}C$ of the accumulated CH_4, while both CO_2 production and $\delta^{13}C$ of CO_2 were hardly affected (Fig. 1). The CO_2 production was probably stable because the sediment is buffered by a high carbonate content. The extent of inhibition of CH_4 production increased only slightly with increase of the concentration of CH_3F from 0.5% to 2%. The $\delta^{13}C$ of the accumulated CH_4 was more negative in the presence than in the absence of CH_3F but decreased only marginally when the CH_3F concentration was increased from 0.5% to 2% (Fig. 1).

In a second experiment, CH_4 again accumulated in the uninhibited control instantaneously and linearly with time, whereas concentrations of acetate stayed constant at about 50 μmol L^{-1} (Fig. 2A). Addition of either 2% (data not shown in Fig. 2A) or 3% (Fig. 2A) CH_3F resulted in partial inhibition of CH_4 production and in accumulation of acetate. The residual rates of CH_4 production were 69% and 65%, respectively. The amounts of accumulated acetate were equivalent to those of CH_4 not produced, showing that acetoclastic methanogenesis was inhibited and was the sole path of acetate consumption. The residual CH_4 accumulating in the presence of CH_3F had a much lower $\delta^{13}C$ than that in the uninhibited control and further decreased with time to values <-70‰, while the δ_{CO_2} stayed constant at about 0‰ (Fig. 2B). The temporal decrease was also seen when calculating the $\delta^{13}C$ in the newly formed CH_4 using Eq. 2. The $\delta^{13}C$ values of total acetate (δ_{ac}) and acetate-methyl ($\delta_{ac\text{-}methyl}$) were higher in the inhibited than in the control incubations (Fig. 2C). Owing to the low acetate concentrations in the uninhibited control, the $\delta^{13}C$ of the acetate-methyl could only be determined by pooling replicate samples. Notably, $\delta_{ac\text{-}methyl}$ was constant with time and similar for both 2% and 3% CH_3F, while δ_{ac}, and thus $\delta_{ac\text{-}carboxyl}$, gradually increased with time of inhibition (Fig. 1C).

In a third experiment, sediment grab samples from the sediment surface (0–5-cm depth) were incubated in the presence of different concentrations of CH_3F (Fig. 3), BES (Fig. 4), and $CHCl_3$ (Fig. 5). We applied BES since it is an analogue of methyl-CoM and is commonly used to inhibit total CH_4 production, and applied $CHCl_3$ since it is believed to inhibit not only methanogens, but also many sulfate reducers and homoacetogens in addition (Chidthai-

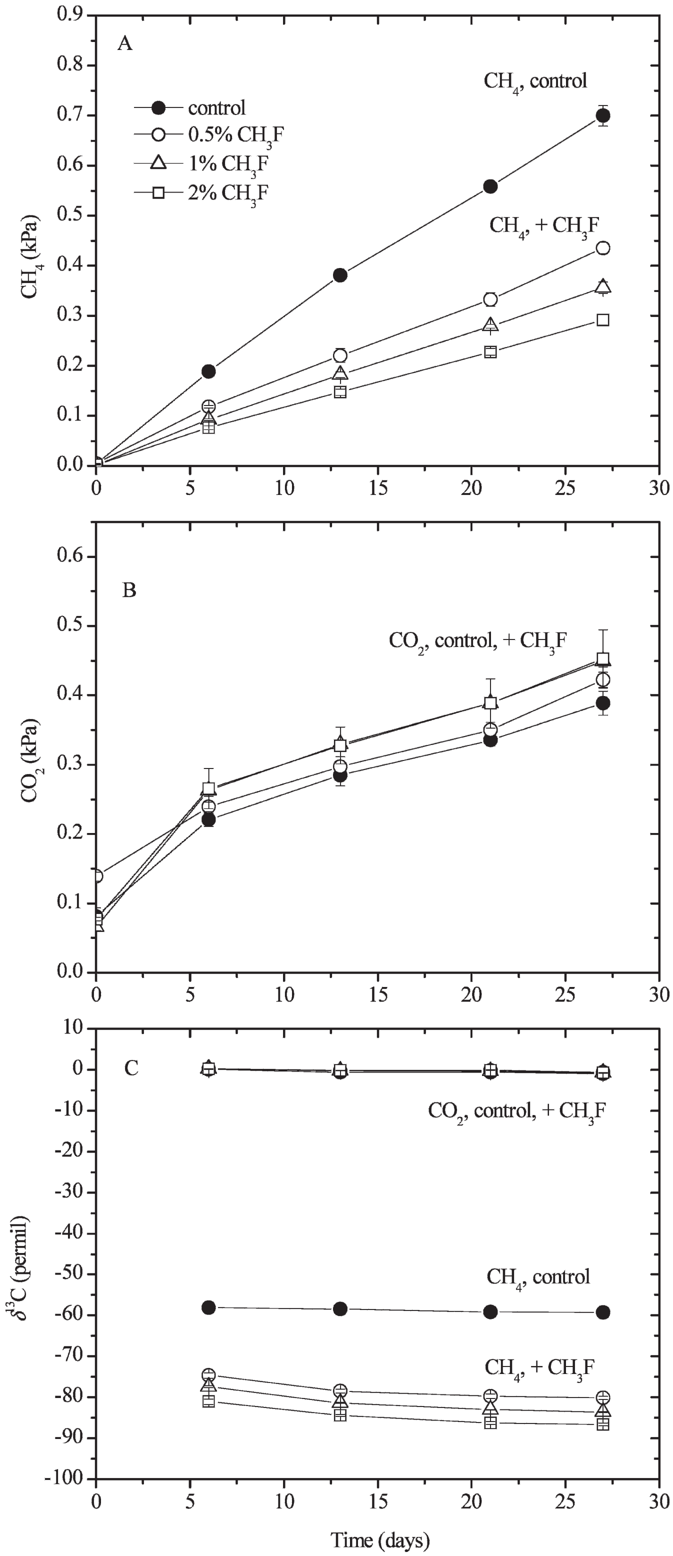

Fig. 1. Production of (A) CH_4, (B) CO_2, and (C) the $\delta^{13}C$ of CH_4 and CO_2 in sediment samples (sampled on 04 Jul 2001, analyzed on 25 Mar 2003) from Lake Dagow incubated anoxically at 10°C in the absence and presence of CH_3F; mean ± SE of $n = 3$.

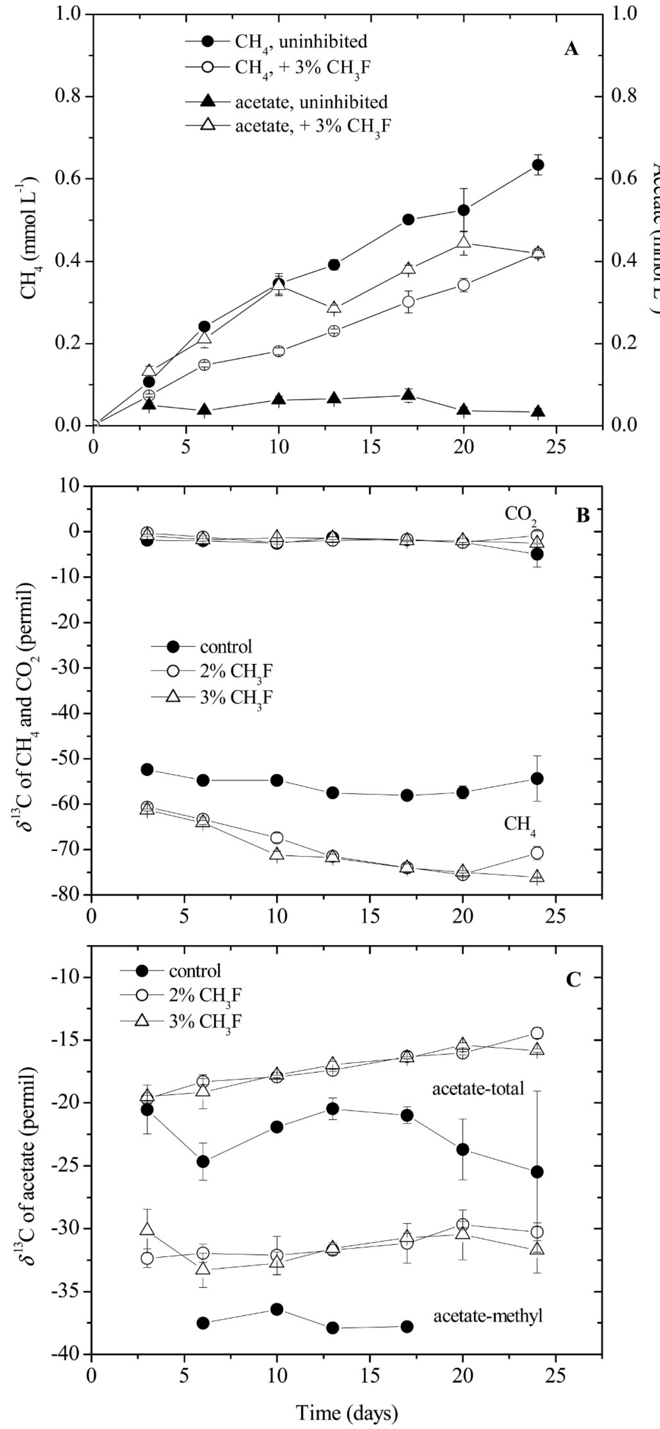

Fig. 2. Production of (A) CH_4 and acetate, (B) $\delta^{13}C$ of CO_2 and CH_4, and (C) $\delta^{13}C$ of total acetate and acetate-methyl in sediment samples (sampled on 17 Aug 2005, analyzed on 23 Aug 2005) from Lake Dagow incubated anoxically at 10°C in the absence and presence of CH_3F. The concentration of CH_4 is given in mmol CH_4 released per liter of sediment; all data are means ±SE of $n = 3$.

song and Conrad 2000; Scholten et al. 2000). The production of CH_4 (Figs. 3A, 4A, 5A) and the $\delta^{13}C$ in the accumulated CH_4 and CO_2 (Figs. 3B, 4B, 5B) were recorded with incubation time, whereas concentrations

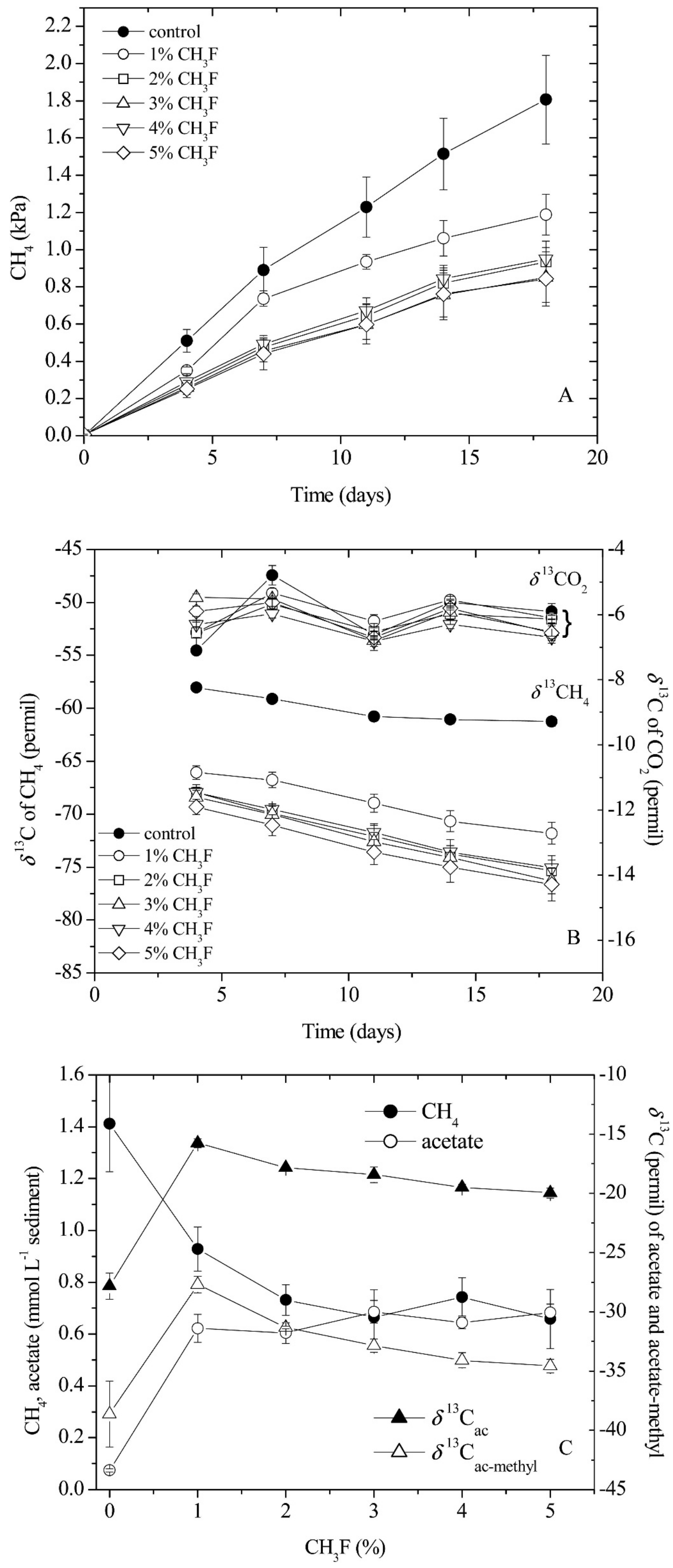

Fig. 3. Production of (A) CH_4 and (B) $\delta^{13}C$ of CO_2 and CH_4 in sediment samples (sampled on 04 Jul 2006, analyzed on 08 Nov 2006) from Lake Dagow incubated anoxically at 10°C in the absence and presence of different concentrations of CH_3F; and (C) CH_4, acetate, and $\delta^{13}C$ of total acetate and acetate-methyl at the end of incubation. The concentration of CH_4 is given in mmol CH_4 released per liter of sediment; all data are means ±SE of $n = 3$.

of acetate and $\delta^{13}C$ of acetate and acetate-methyl were only determined at the end of incubation and then plotted as a function of inhibitor concentration (Figs. 3C, 4C, 5C).

Methyl fluoride inhibited CH_4 production only partially and reached maximum inhibition already at 2% CH_3F as observed in the other two experiments (Fig. 3A). Compared with the uninhibited control, addition of ≥2% CH_3F again resulted in much lower δ_{CH_4}, while δ_{CO_2} was not affected (Fig. 3B). Again, δ_{CH_4} decreased with incubation time, both for accumulated CH_4 (Fig. 3B) and for newly formed CH_4 (data not shown). The amounts of acetate accumulated were stoichiometric to the amount of CH_4 not produced (Fig. 3C), indicating that CH_3F specifically inhibited acetoclastic methanogenesis. Inhibition resulted in higher $\delta^{13}C$ of total acetate and acetate-methyl, but not so much at higher than at lower CH_3F concentrations (Fig. 3C; 1% to 5% CH_3F).

Addition of BES eventually resulted in the complete inhibition of CH_4 production (Fig. 4A). However, this inhibition was not instantaneous and only effective at high (30 mmol L^{-1}) BES concentrations. The $\delta^{13}C$ of the accumulated CH_4 was on the same range for both the inhibited and the uninhibited experiment but increased with incubation time while CH_4 production became increasingly inhibited (Fig. 4B). The δ_{CO_2} in the inhibited incubations was rather constant with time but was slightly higher than in the control and increased with increasing BES concentration (Fig. 4B). This increase was possibly due to a decrease of the pH (from 7.5 to 7.1) by the addition of BES and consequently mobilization of relatively heavy carbonate C. This mobilization was also seen by CO_2 release, which increased with increasing BES concentration (data not shown). Addition of BES also resulted in accumulation of acetate, but as expected the amounts were less than the stoichiometric equivalent of the CH_4 not produced (Fig. 4C), showing that BES also inhibited hydrogenotrophic in addition to acetoclastic methanogenesis. The $\delta^{13}C$ of acetate-methyl was similar to that during CH_3F inhibition and again decreased with increasing inhibitor concentrations (Fig. 4C, 5 to 30 mmol L^{-1} BES). The $\delta^{13}C$ of total acetate, on the other hand, was only slightly higher in the presence of BES than in the uninhibited control.

Addition of $CHCl_3$ resulted in instantaneous and almost complete inhibition of CH_4 production, even at the lowest (50 μmol L^{-1}) $CHCl_3$ concentration (Fig. 5A). The $\delta^{13}C$ of the little CH_4 accumulated in the presence of $CHCl_3$ was slightly lower than that of the uninhibited control and the δ_{CO_2} in the gas phase was slightly higher, but both stayed constant with time (Fig. 5B). The slightly higher δ_{CO_2} was not caused by decrease of pH, which stayed constant, but was possibly due to partial inhibition of conversion of organic matter (with low $\delta^{13}C$) to CO_2 by $CHCl_3$. Acetate accumulated in the presence of $CHCl_3$, but the amount decreased with increasing $CHCl_3$ concentration, indicating inhibition of acetate-producing processes (Fig. 5C). The $\delta^{13}C$ of the acetate-methyl was similar as in the experiments with CH_3F and BES and again slightly higher than in the uninhibited control. However, the $\delta^{13}C$ of total acetate was

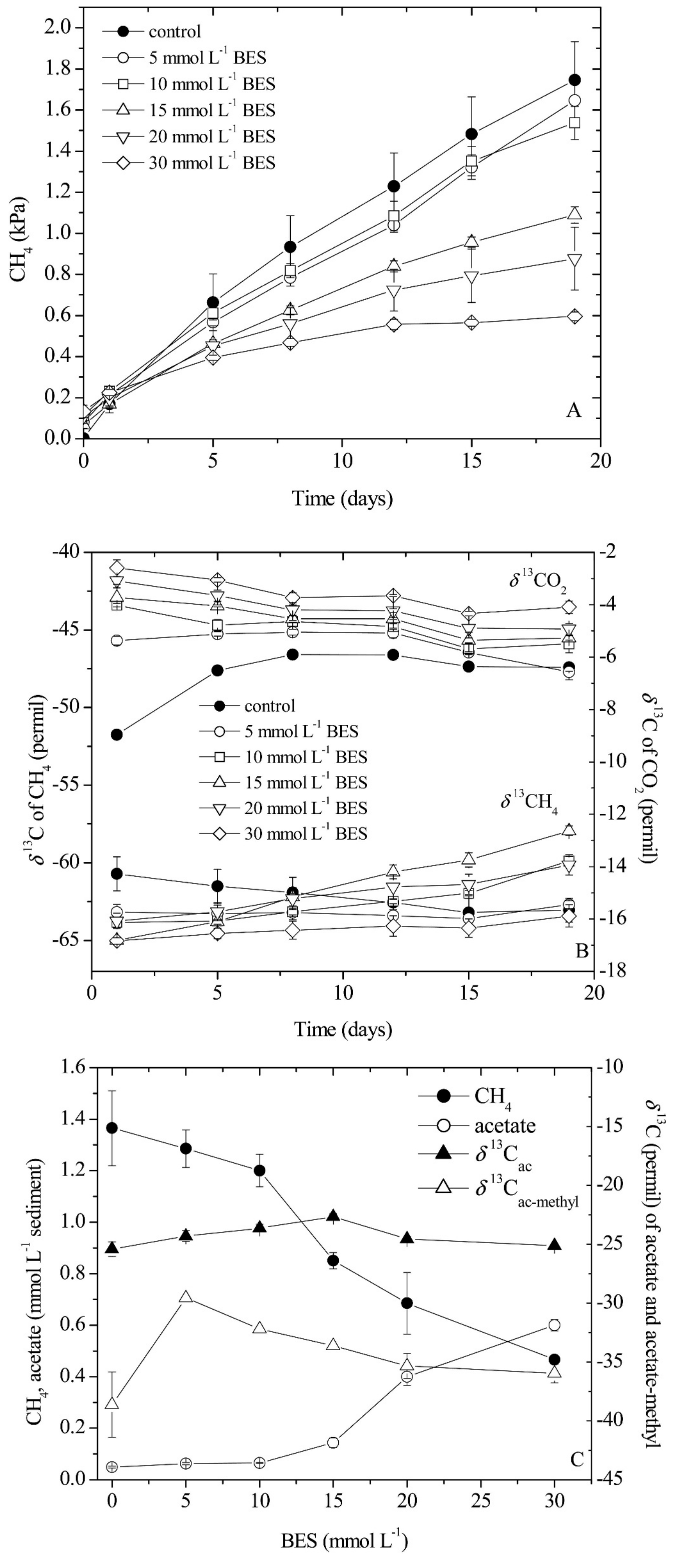

Fig. 4. Production of (A) CH_4 and (B) $\delta^{13}C$ of CO_2 and CH_4 in sediment samples (sampled on 04 Jul 2006, analyzed on 08 Nov 2006) from Lake Dagow incubated anoxically at 10°C in the absence and presence of different concentrations of BES; and (C) CH_4, acetate, and $\delta^{13}C$ of total acetate and acetate-methyl at the end of incubation. The concentration of CH_4 is given in mmol CH_4 released per liter of sediment; all data are means ±SE of n = 3.

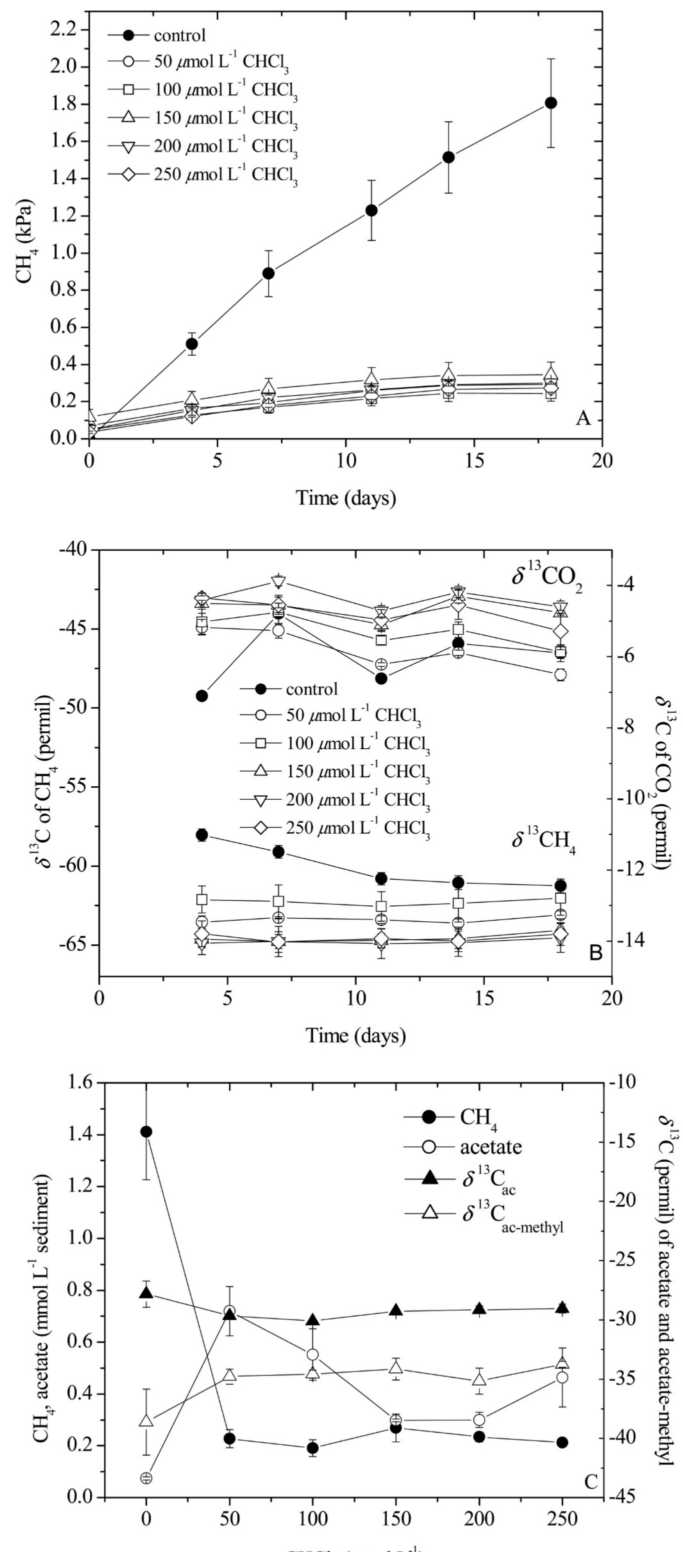

Fig. 5. Production of (A) CH_4 and (B) $\delta^{13}C$ of CO_2 and CH_4 in sediment samples (sampled on 04 Jul 2006, analyzed on 08 Nov 2006) from Lake Dagow incubated anoxically at 10°C in the absence and presence of different concentrations of chloroform; and (C) CH_4, acetate, and $\delta^{13}C$ of total acetate and acetate-methyl at the end of incubation. The concentration of CH_4 is given in mmol CH_4 released per liter of sediment; all data are means ±SE of n = 3.

less negative and similar as in the inhibited and uninhibited samples (Fig. 5C).

Vertical profiles of Lake Dagow sediment—Sediment cores were retrieved from Lake Dagow in May and Aug 2003 and processed immediately. In May, stratification of the water column had just completed, and O_2 in the hypolimnion had already decreased to $<1\%$ saturation. Accumulation of CH_4 and $\delta^{13}C$ of CH_4 and CO_2 was followed with time using sediment samples from different depths of triplicate sediment cores. The primary results are not shown but were similar to those shown in Fig. 1. Production of CH_4 was highest in the upper 3-cm sediment layer, being about 200 to 350 nmol h^{-1} g^{-1} dry weight. Production rates of CH_4 decreased with depth reaching a constant value of about 10 nmol h^{-1} g^{-1} below 10-cm depth. In the presence of CH_3F, CH_4 production rates were generally lower, but rates were similar with either 2% or 3% CH_3F. The residual CH_4 production rates increased with depth from about 20–30% at the surface to 50–60% at 10–15-cm depth and decreased again to about 35% at 15–20-cm depth (data not shown).

The $\delta^{13}C$ of CH_4 and CO_2 was also measured in the slurries of the sediment sections. Similarly as in the experiments shown in Fig. 1, values of δ_{CH_4} rapidly reached a constant value and did not further decrease with time. The $\delta^{13}C$ of newly produced CH_4 was calculated from the accumulated CH_4 using Eq. 2. The $\delta^{13}C$ of CH_4 and CO_2 are plotted in Fig. 6. The diagram also shows the isolines of different apparent fractionation factors (α_{app}) for CO_2 reduction to CH_4 as calculated by Eq. 3. The $\delta^{13}CO_2$ generally increased with sediment depth from about −14‰ to 0‰ (Fig. 6). The $\delta^{13}CH_4$ produced in the presence of CH_3F increased with depth from about −100‰ to −80‰, which is equivalent to a α_{CO_2/CH_4} of about 1.090–1.095 (Fig. 6). The CH_4 produced in the uninhibited control, on the other hand showed a rather constant $\delta^{13}CH_4$ of about −65‰, with a slightly decreasing trend with increasing sediment depth, so that α_{app} ranged between of 1.05 to 1.07 (Fig. 6).

Figure 7 summarizes the $\delta^{13}C$ values in the vertical profiles of uninhibited sediment for CH_4, CO_2, organic carbon, total C, and carbonate. The latter was calculated from the former two by $\delta_{carb} = \delta_{tot} - \delta_{org}$. Values of δ_{carb} slightly increased with sediment depth from 5‰ to 9‰, whereas δ_{org} was constant at a low −30‰ (Fig. 7). The difference between δ_{tot} and δ_{org} reflects the high carbonate content of the sediment, which had a much higher $\delta^{13}C$ than the organic matter.

The in situ concentrations of acetate were below the detection limit of 5 μmol L^{-1}, so that the $\delta^{13}C$ of acetate could not be measured directly. Therefore, the acetate that had accumulated in the incubation experiments of the depth sections in the presence of 2% and 3% CH_3F was analyzed at the end of the incubation. The $\delta^{13}C$ of the accumulated acetate-methyl was relatively low and seemed to increase slightly with depth from −38‰ to −30‰ at 0–3 cm to −33‰ to −28‰ at 15–20 cm (Fig. 7). The δ_{ac}, on the other hand, was higher at about −22‰ to −19‰ and stayed fairly constant with depth (Fig. 7B).

Discussion

Conversion of sediment organic carbon to methane—The δ_{org} found in the sediment of Lake Dagow (–31‰) is characteristic for C3 photosynthesis (−27‰) (O'Leary 1981). The light organic carbon is not unusual for eutrophic lakes and is probably caused by sedimentation of microbial biomass (Hollander and Smith 2001; Teranes and Bernasconi 2005). Lake Dagow sediment is characterized by a relatively high (23–39%) content of calcium carbonate, which has built up during autochthonic and organismic calcite precipitation in the summer months (Koschel et al. 1990). The carbonate was isotopically heavy (5–9‰) compared with the organic matter reflecting isotope fractionation during photosynthetic biomass production and heterotrophic processes within the food web and during terminal mineralization as methanogenesis (Teranes and Bernasconi 2005).

In all experiments, CH_4 production was instantaneous. This result was expected, since the sediment samples were taken during a period where the lake was stratified and had an anoxic hypolimnion. Hence, inorganic electron acceptors, such as nitrate, sulfate, or ferric iron, were depleted and reduction of CO_2 and of methyl groups to CH_4 were the only conceivable redox processes. The exclusivity of methanogenesis has previously been shown for Lake Dagow sediment by measuring the turnover of [2-^{14}C] acetate, demonstrating that the methyl group of acetate was converted to CH_4 rather than CO_2 (Glissmann et al. 2004). Methane production was completely inhibited when applying BES or $CHCl_3$. However, CH_4 production was only partially inhibited by CH_3F. Methyl fluoride (CH_3F) was reported to be a specific inhibitor of acetoclastic methanogens, while hydrogenotrophic methanogens are not affected (Janssen and Frenzel 1997). Our results are consistent with this interpretation, since acetate accumulated stoichiometrically with the CH_4 inhibited.

The δ_{CH_4} in uninhibited Lake Dagow sediment was about −60‰ to −55‰ and either stayed constant (Figs. 1, 2) or slightly decreased with time (Figs. 3–5). At the same time, δ_{CO_2} was about −7‰ to 0‰. This rather high value is intermediate between the $\delta^{13}C$ of carbonate and organic carbon and is most probably the result of mixing of CO_2 produced from (1) dissolution of carbonate, (2) degradation of organic matter, and (3) fractionation of CO_2 during hydrogenotrophic methanogenesis. Thus, values of $\delta^{13}C$ increased with decreasing pH (caused by BES) as a result of increased dissolution of heavy carbonate carbon (Fig. 4) but also increased when respiration of light organic carbon was inhibited by $CHCl_3$ (Fig. 5).

The measured values of δ_{CH_4} and δ_{CO_2} give an apparent fractionation factor (Eq. 3) of $\alpha_{app} \approx 1.050$–1.063 or $\varepsilon_{app} \approx$ −63‰ to −50‰ (Figs. 8, 9). This α_{app} is intermediate between the α values that are typically found for either hydrogenotrophic or acetoclastic methanogens (*see* introduction), thus indicating that both methanogenic paths were operating in the uninhibited sediment. The $\delta^{13}C$ of total acetate (without inhibitor) was between −28‰ and −20‰ (Figs. 2–5) and exhibited no apparent tendency over time (Fig. 2).

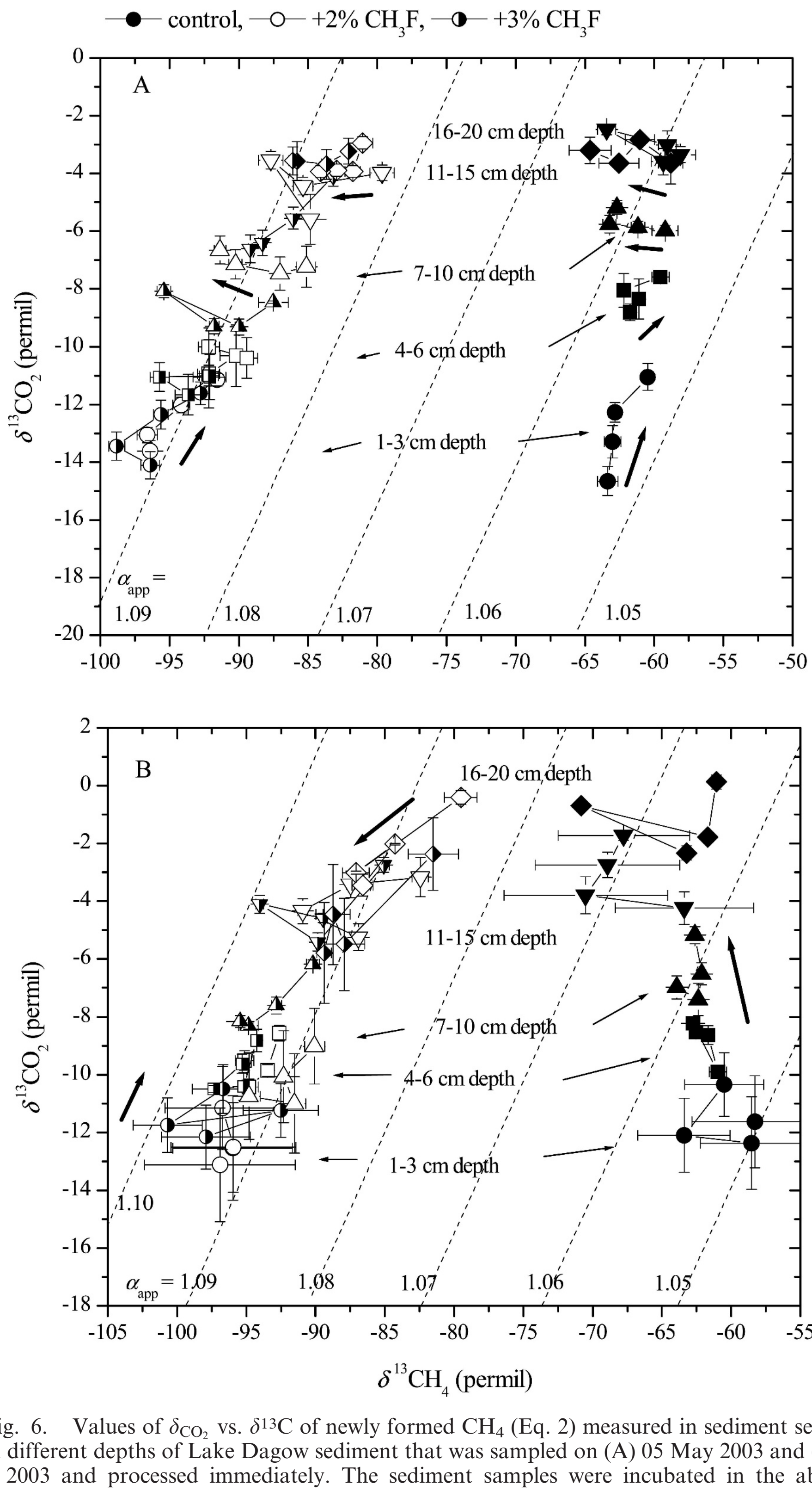

Fig. 6. Values of δ_{CO_2} vs. $\delta^{13}C$ of newly formed CH_4 (Eq. 2) measured in sediment sections from different depths of Lake Dagow sediment that was sampled on (A) 05 May 2003 and (B) 04 Aug 2003 and processed immediately. The sediment samples were incubated in the absence (closed symbols) or presence of 2% (open symbols) or 3% (half-open symbols) CH_3F. The thick arrows indicate the time course of incubation (up to 18–25 d). The dotted isolines show the values of different α_{app} (Eq. 3); mean $\pm$ SE of n = 3.

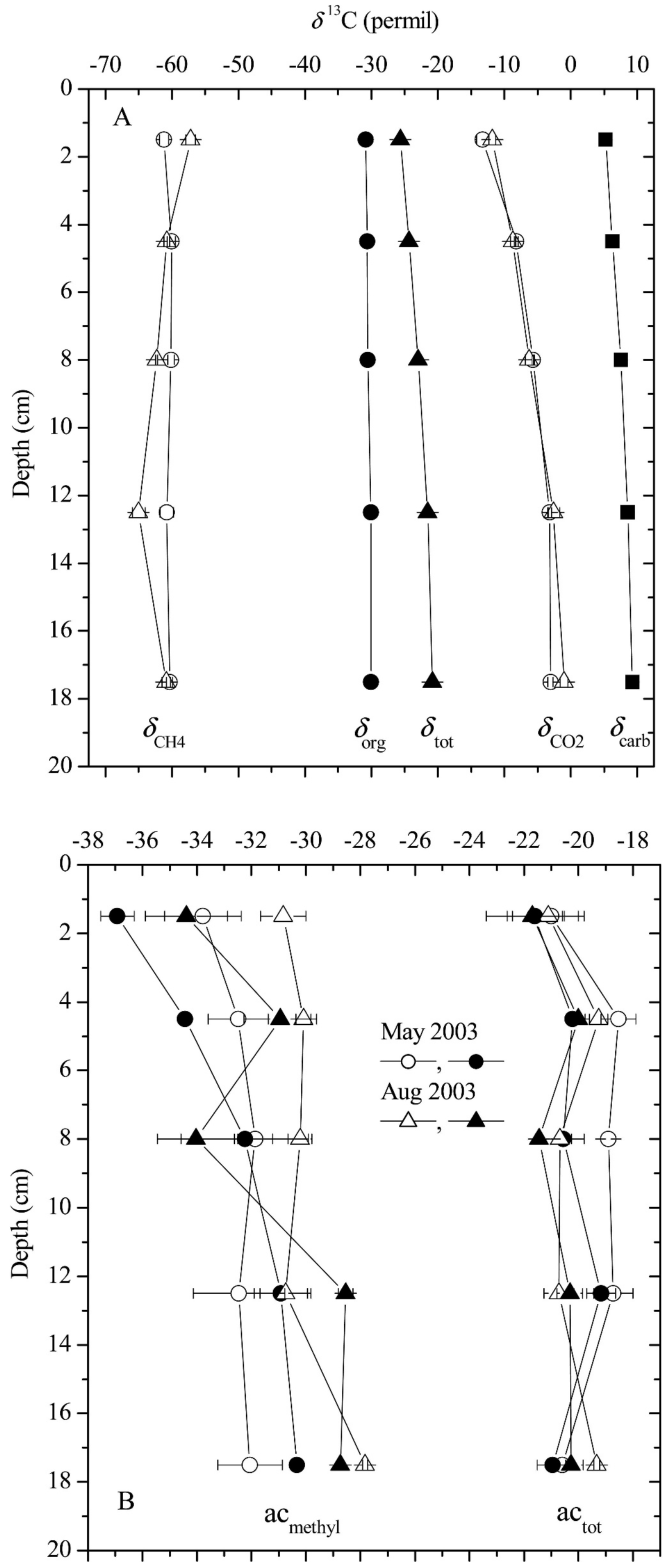

Fig. 7. Vertical profiles of (A) $\delta^{13}C$ of CO_2, CH_4, total carbon, organic carbon, and carbonate carbon, and (B) $\delta^{13}C$ of total acetate and acetate-methyl in profundal sediment of Lake Dagow. Data of δ_{tot}, δ_{org}, and δ_{carb} were from sediment samples taken on 12 Jun 2003, the other data were from samples taken on 05 May 2003 (circles) or 04 Aug 2003 (triangles). The δ_{CO_2} and δ_{CH_4} data were averaged over 18–25 d of incubation, the acetate data were obtained after incubation of the sediment in the presence of 2% (open symbols) and 3% (closed symbols) CH_3F; mean ±SE of n = 3.

Fractionation during hydrogenotrophic methanogenesis—Addition of CH_3F partially inhibited methanogenesis. Inhibition eventually became saturated when CH_3F concentrations were further increased. Addition of CH_3F also resulted in a decrease of $\delta^{13}CH_4$, and this decrease eventually leveled off when CH_3F concentrations were further increased (Figs. 1–5). The observations that increase of CH_3F concentration did not result in a change of the $\delta^{13}CH_4$ indicated that acetoclastic methanogenesis was inhibited completely and that the $\delta^{13}CH_4$ then represented the $\delta^{13}C$ (δ_{mc}) of the CH_4 produced from H_2 plus CO_2. If we accept that acetoclastic methanogenesis was completely inhibited in the presence of CH_3F, we can use the resulting values of δ_{CH_4} and δ_{CO_2} to calculate α_{CO_2/CH_4}, the fractionation factor for reduction of CO_2 to CH_4. If acetoclastic methanogenesis is not operating, α_{CO_2/CH_4} should be equal to α_{app}, i.e.,

$$\alpha_{CO_2/CH_4} = (\delta_{CO_2} + 1000)/(\delta_{mc} + 1000) \qquad (5)$$

In some incubations values of δ_{CH_4} continuously decreased with incubation time (Figs. 2, 3), but not in all (Figs. 1, 6). Notably, in CH_3F incubations of surface sediment, which had been stored for 8 months, δ_{CH_4} did not decrease strongly (Fig. 1). Also in CH_3F incubations of sections of freshly taken sediment cores, δ_{CH_4} did not decrease strongly (Fig. 6). Furthermore, δ_{CH_4} of accumulated CH_4 in these experiments (Figs. 1, 6) reached values of −95‰, whereas those in the former (Figs. 2, 3) reached only −75‰, indicating that a new steady state upon inhibition with CH_3F had probably not yet been reached in the former experiments. The most parsimonious explanation is that fractionation of CO_2 reduction became larger in newly formed CH_4 since the isotopic fractionation factor increased with time. This was indeed seen for the respective incubations when α_{CO_2/CH_4} was calculated (Figs. 8, 9). In some of the experiments with Lake Dagow sediment α_{CO_2/CH_4} increased with time from about 1.070 to 1.095 (Fig. 8B). In other experiments with surface sediment of Lake Dagow, on the other hand, α_{CO_2/CH_4} was constant right from the beginning at about 1.090–1.098 (Fig. 8A). Also in the vertical profiles of Lake Dagow sediment α_{CO_2/CH_4} showed time-constant values (1.090–1.095) at the surface layers decreasing with depth to about 1.085–1.090 (Fig. 9). Notably, values of α_{CO_2/CH_4} were apparently not constant with depth, so that explicit determination of α_{CO_2/CH_4} was warranted.

We hypothesize that the time-dependent behavior of α_{CO_2/CH_4} was primarily influenced by the energetic conditions for hydrogenotrophic methanogenesis, which strongly affect isotope fractionation (Valentine et al. 2004; Penning et al. 2005). Thus, α_{CO_2/CH_4} increases as the Gibbs free energy (ΔG) of hydrogenotrophic methanogenesis becomes less negative, so that high values of α_{CO_2/CH_4} indicate more

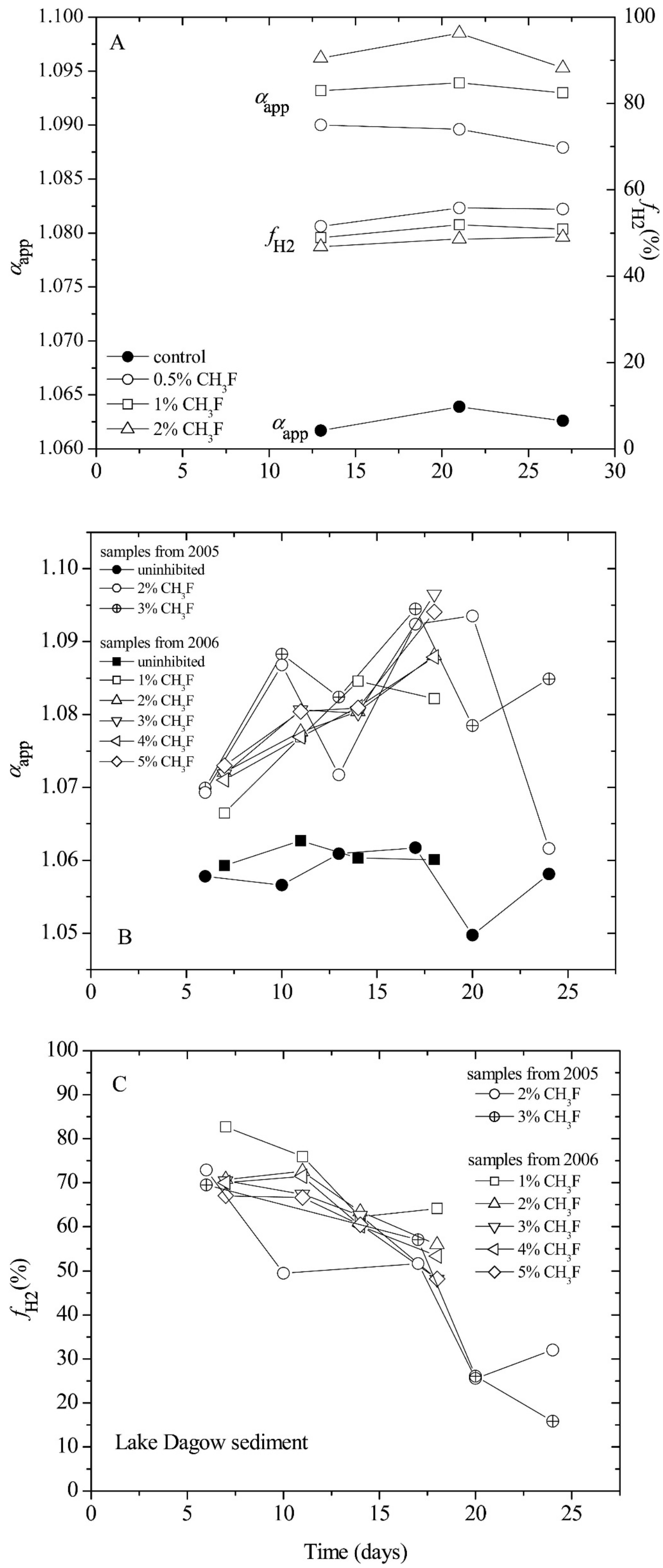

Fig. 8. Change of apparent fractionation factors (Eq. 3) and percentage contributions (f_{H_2}) of hydrogenotrophic methanogenesis to total CH_4 production (Eq. 4) during incubation of sediment samples. In (A) values of α_{app} and f_{H_2} were calculated from the mean values shown in Fig. 1; in (B) values of α_{app} were calculated from the mean values shown in Figs. 2 and 3; and in (C) values of f_{H_2} were calculated from the mean values shown in Figs. 2 and 3.

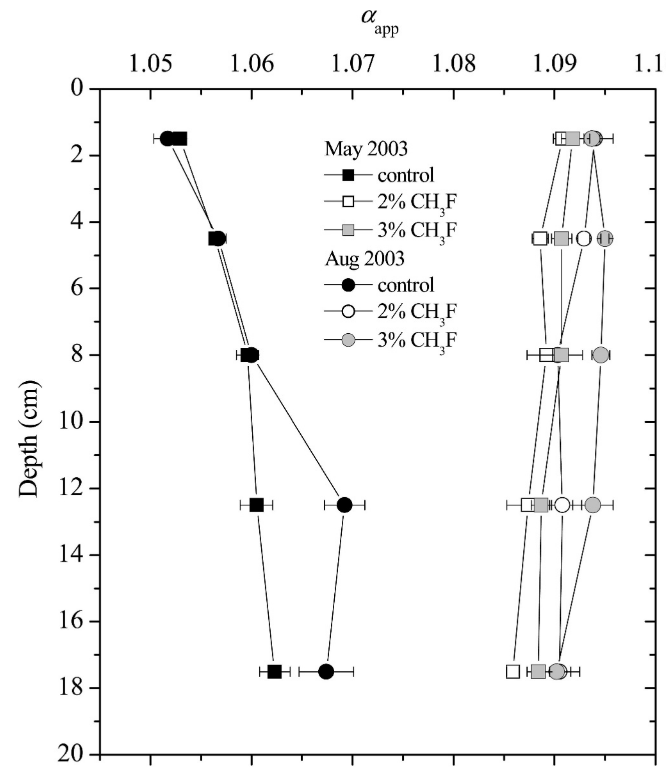

Fig. 9. Vertical profiles of α_{app} (Eq. 2) determined from the newly produced CH_4 in the absence (closed symbols) and presence (open symbols) of 2% and 3% CH_3F determined in the profundal sediment of Lake Dagow sampled on 05 May and 04 Aug 2003. Values of α_{app} determined in the presence of CH_3F are equivalent to α_{CO_2/CH_4}; mean ± SE of $n = 3$.

limited energetic conditions than low values (Penning et al. 2005). The mechanistic reasons for this behavior are discussed by Valentine et al. (2004) and Penning et al. (2005). Collectively, our experiments indicate that energetic conditions of hydrogenotrophic methanogenesis in sediment of Lake Dagow must be relatively restricted, since values of α_{CO_2/CH_4} were on the order of 1.085–1.095, equivalent to values of ΔG being on the order of -20 to 0 kJ mol^{-1} (Penning et al. 2005).

When CH_4 production was completely inhibited by chloroform (Fig. 5), the little residual CH_4 produced generally had slightly lower $\delta^{13}C$ than the CH_4 in the untreated control. We assume that the relatively low δ_{CH_4} in the chloroform treatments were caused by hydrogenotrophic methanogenesis becoming less rapidly inhibited than acetotrophic methanogenesis. In the BES treatments (Fig. 4), the δ_{CH_4} was similarly low at the beginning of incubation, but in some of the BES incubations δ_{CH_4} increased with time, in particular in those with 10–20 mmol L^{-1} BES, in which CH_4 production was only partially inhibited. Since final acetate concentrations in the 5–15 mmol L^{-1} BES treatments were relatively low, some acetoclastic methanogenesis might still have been operating while hydrogenotrophic methanogenesis was already completely inhibited by BES, thus resulting in a gradual increase of $\delta^{13}CH_4$.

Fractionation of acetate carbon—In Lake Dagow sediment acetate continuously accumulated upon addition of CH_3F, while $\delta_{\text{ac-methyl}}$ of the accumulating acetate stayed at a constant value (Fig. 2). We assume that these values are the signature of the fermentatively produced acetate-methyl, which were about −33‰ to −30‰. $\delta_{\text{ac-methyl}}$ was thus similar to the δ_{org}, indicating that there was no significant fractionation during the fermentation of organic matter to acetate. This is not unexpected given the fact that fermentation of glucose to acetate was found to have only a small isotope effect (Blair et al. 1985; Penning and Conrad 2006). However, since we only know the $\delta^{13}C$ of total organic matter but not the $\delta^{13}C$ of the precursors of acetate, there might be fractionation going on that is masked by the background of undegraded organic carbon.

Based on our knowledge of fractionation during fermentation (Blair et al. 1985), we may expect $\delta^{13}C$ values higher by 24‰ in the carboxyl group than in the methyl group of acetate or differences of $\delta_{\text{ac}} - \delta_{\text{ac-methyl}}$ of about 12‰. For Lake Dagow sediment, we found an average value of $\delta_{\text{ac}} - \delta_{\text{ac-methyl}}$ of >12‰ in the presence of CH_3F, which slightly increased with time to 16‰ (Fig. 2). The temporal increase may be due to an enzymatically catalyzed exchange reaction of the carboxyl group with the CO_2 in the environment ($\delta_{CO_2} \approx 0\%$), in particular since acetate accumulated during incubation. Such exchange reactions have been reported for methanogenic environments (Gelwicks et al. 1994; DeGraaf et al. 1996). Note that δ_{ac} in the inhibited incubations increased with time relative to that in the control (Fig. 2), indicating that exchange was mainly active when acetate was not consumed but accumulated with time.

Interestingly, the acetate in the uninhibited control incubations exhibited lower values of $\delta_{\text{ac-methyl}}$ (−38‰ to −37‰) than in the inhibited incubations (−34‰ to −30‰) (Fig. 2). The same was true for total acetate (δ_{ac}), which also showed more negative values in the absence (−25‰ to −21‰) than in the presence of CH_3F (−19‰ to −15‰) (Fig. 2). The similarity of the results obtained for δ_{ac} rule out a bias during pyrolysis of acetate for determination of $\delta_{\text{ac-methyl}}$. Similar results were obtained when using different CH_3F concentrations (Fig. 3) or when total CH_4 production was inhibited with either BES (Fig. 4) or $CHCl_3$ (Fig. 5), albeit in the BES and $CHCl_3$ treatments only $\delta_{\text{ac-methyl}}$ was lower in the control than in the inhibitor treatments, and δ_{ac} was fairly constant regardless of the addition of different inhibitor concentrations. These results were unexpected, since consumption of acetate in the absence of an inhibitor should result in the preferred consumption of ^{12}C-acetate so that δ_{ac} and $\delta_{\text{ac-methyl}}$ should increase rather than decrease. In fact, such a normal fractionation behavior has previously been observed in methanogenic rice field soil (Penning and Conrad 2007) and rice root preparations (Penning et al. 2006*b*), where the data were consistent with a fractionation factor $\alpha_{\text{ac-methyl}/CH_4} = 1.01 - 1.02$ that is characteristic for acetoclastic methanogens. Fractionation of acetate carbon is to be expected in systems in which acetate production is transiently high, such as in rice fields. In the methanogenic sediment of oligotrophic Lake Stechlin, δ_{ac} and $\delta_{\text{ac-methyl}}$ were similar in the presence or absence of CH_3F (Conrad et al. 2007). Such data indicate that there is no fractionation ($\alpha_{\text{ac-methyl}/CH_4} = 1$) during acetoclastic methanogenesis. This happens if all the acetate produced during fermentation is subsequently consumed. Such tight coupling between production and consumption is plausible if acetate production rates limit acetoclastic methanogenesis, which is to be expected in systems that have attained steady state, such as lake sediments. However, our data in Lake Dagow sediment indicate fractionation factors <1, which cannot be explained by a kinetic isotope effect.

Unfortunately, literature data on δ_{ac} and $\delta_{\text{ac-methyl}}$ are scarce. Nevertheless, we may hypothesize two conceivable reasons for the nonnormal isotope fractionation observed, which was rather small (≈5‰) but systematic. We assume that it may either be an effect of the inhibitors on fermentative acetate production or an unusual fractionation during acetoclastic methanogenesis or both. For example, if CH_3F would not only inhibit acetoclastic methanogens, but also a few species of the many potential acetate-producing microorganisms, the isotope fractionation during acetate production might be affected. Application of different inhibitors produced slightly different effects, which seemed to be stronger for the more specific than for the unselective inhibitors (i.e., CH_3F > BES > $CHCl_3$). Also lower concentrations of the inhibitors seemed to have a stronger effect than larger concentrations. These observations are consistent with our speculation, since unspecific inhibitors and higher inhibitor concentrations would affect all microbial species equally, whereas specific inhibitors and low inhibitor concentrations would affect only a few of these microbes. In our example with Lake Dagow sediment, these would be microbes that produce acetate with a slightly more negative $\delta^{13}C$, so that δ_{ac} and $\delta_{\text{ac-methyl}}$ would increase upon addition of inhibitor, as observed.

Alternatively, we may assume that the observed difference in $\delta_{\text{ac-methyl}}$ in the presence and absence of inhibitor was due to an equilibrium isotope effect. Recently, we found that some species of acetate-using sulfate-reducing bacteria (those using the tricarbonic acid cycle for acetate catabolism) exhibit positive enrichment factors, which seem to be due to a lack of a kinetic isotope effect during acetate catabolism plus an equilibrium isotope effect caused by the protonation of acetate (Goevert and Conrad 2008). Protonation of acetate is necessary for transport over the lipophilic cell membrane. This is also the case for acetoclastic methanogens, so that such an equilibrium isotope effect may also be operative in these microbes. However, the rather small equilibrium isotope effect is in methanogens normally masked by the much larger kinetic isotope effect. Only if acetoclastic methanogens were limited by acetate (as is likely the case in Lake Dagow sediment) would the kinetic isotope effect become zero while the equilibrium isotope effect would still be expressed.

Independently of whether hypothesis one or two, or both, are true, the effect on the isotopic composition of acetate-methyl was rather small (≈ 5‰) compared with the potential kinetic isotope effect during acetoclastic methanogenesis (ε_{ac/CH_4} of −30‰ to −10‰). Therefore, we make only a rather small error when using the measured values of $\delta_{ac\text{-}methyl}$ for calculation of the relative contribution of hydrogenotrophic and acetoclastic methanogenesis to total CH_4 production.

Relative contribution of hydrogenotrophic and acetoclastic methanogenesis to total CH_4 *production*—Since inhibition of methanogenesis and decrease of δ_{CH_4} was eventually saturated by increasing concentrations of CH_3F, we may assume that inhibition of acetoclastic methanogenesis by CH_3F was complete. Possibly, a small part of the hydrogenotrophic methanogenesis became in addition inhibited, as observed in previous experiments with rice field soil (Conrad and Klose 1999). Therefore, the direct calculation of the relative contribution of hydrogenotrophic methanogenesis (f_{H_2}) from the residual activity after CH_3F inhibition is probably biased. It is more robust to calculate f_{H_2} from the isotopic mass balance, which should not or little be affected when part of the hydrogenotrophic methanogenesis is inhibited (Eq. 5). We need three different input variables, i.e., δ_{CH_4}, the $\delta^{13}C$ of the newly produced total CH_4 in the uninhibited control; δ_{mc}, the $\delta^{13}C$ of the newly produced hydrogenotrophic CH_4, which is equal to the $\delta^{13}C$ of the newly produced CH_4 in the inhibited incubation; and δ_{ma}, the $\delta^{13}C$ of the newly produced acetoclastic CH_4. The value of δ_{ma} depends on $\delta_{ac\text{-}methyl}$ and the fractionation factor $\varepsilon_{ac\text{-}methyl/CH_4}$, i.e.,

$$\delta_{ma} = \delta_{ac\text{-}methyl} + \varepsilon_{ac\text{-}methyl/CH_4} \tag{6}$$

As discussed above, $\varepsilon_{ac\text{-}methyl/CH_4}$ is likely close to zero if we assume that $\delta_{ac\text{-}methyl}$ is represented by the $\delta^{13}C$ of the acetate accumulated in the presence of CH_3F ($\delta_{ac\text{-}methyl\text{-}CH_3F}$). However, if we assume that the $\delta^{13}C$ of the fermentatively produced acetate was biased by CH_3F inhibition, $\delta_{ac\text{-}methyl}$ would probably be better represented by the acetate in the uninhibited controls ($\delta_{ac\text{-}methyl\text{-}control}$). Hence, δ_{ma} might actually be by 5‰ lower than the $\delta_{ac\text{-}methyl\text{-}CH_3F}$.

As a first approximation we assumed $\delta_{ma} = \delta_{ac\text{-}methyl\text{-}CH_3F}$ and used the measured values of δ_{CH_4} and δ_{mc} for calculation of f_{H_2} in Lake Dagow sediment. In one of the sediment samples (Fig. 1) f_{H_2} stayed constant at 50–55% (Fig. 8A). In another sediment sample (Fig. 2), f_{H_2} decreased from initially about 75% to 16–32% in the end (Fig. 8C). The final values of f_{H_2} of about 16–32% are consistent with the biochemistry of theoretical degradation of organic matter expecting a contribution of <33% at steady state (Conrad 1999) and are consistent with earlier experiments on grab samples of Lake Dagow sediment using radiotracers (Glissmann et al. 2004). In a third sediment sample (Fig. 3), f_{H_2} also decreased from about 70–85% but only to 50–65% as the incubation was then finished (Fig. 8C). The decrease of f_{H_2} was mainly caused by the decrease in δ_{mc} and the increase of α_{CO_2/CH_4} (Fig. 8B). The increase of α_{CO_2/CH_4} may be due to change in energetic conditions as discussed above, but it is unclear why it was not expressed in the sediment sample obtained in 2001. It was also not expressed in the vertical sediment profiles (Fig. 6). Note that the results would not be strongly affected when using values of δ_{ma} that are by 5‰ more negative than the $\delta_{ac\text{-}methyl\text{-}CH_3F}$. In this case, f_{H_2} would be by 4–5% smaller, but the temporal trends would be the same.

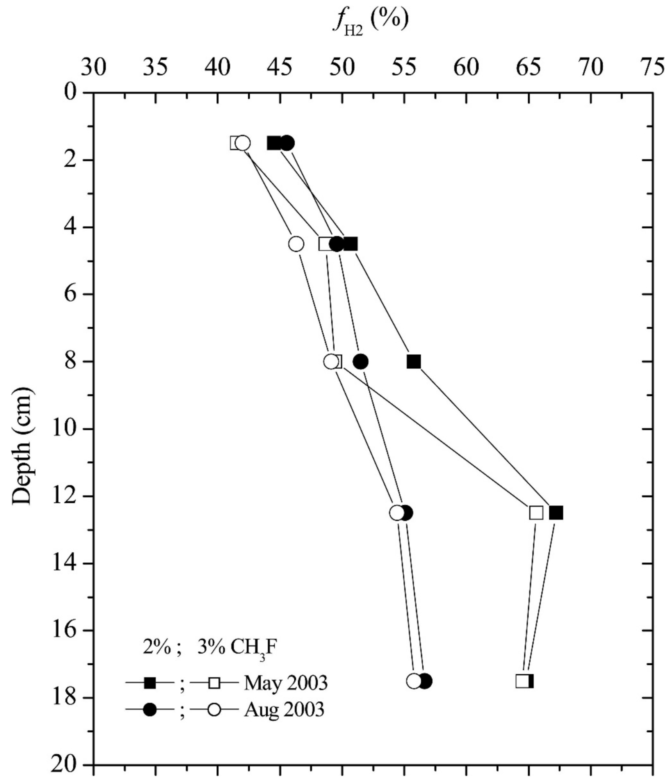

Fig. 10. Vertical profiles of the percentage contributions (f_{H_2}) of hydrogenotrophic methanogenesis to total CH_4 production in the profundal sediment of Lake Dagow sampled on 05 May and 04 Aug 2003. f_{H_2} was calculated from Eq. 4 using the $\delta^{13}C$ of newly formed CH_4 and assuming $\delta_{ma} = \delta_{ac-methyl-CH_3F}$.

For the vertical sediment profile in Lake Dagow we also used the values of δ_{CH_4} and δ_{mc}, which stayed fairly constant over time and averaged them. In addition, we again used $\delta_{ma} = \delta_{ac\text{-}methyl\text{-}CH_3F}$. Vertical profiles of f_{H_2} were also calculated from the data obtained with the two different CH_3F concentrations and from two sampling times (Fig. 6). They consistently showed that f_{H_2} was increasing with sediment depth from about 40–45% at the surface to about 55–67% at 10–20-cm depth (Fig. 10). The increase of f_{H_2} with depth was mainly caused by the decrease in α_{CO_2/CH_4} (Fig. 9) and the increase in δ_{CO_2} (Fig. 7), resulting in increase of δ_{mc} with depth, while $\delta_{ma} = \delta_{ac\text{-}methyl\text{-}CH_3F}$ (Fig. 7) increased only very slightly. The increase of f_{H_2} with sediment depth was paralleled by an increase in the proportion of hydrogenotrophic vs. acetoclastic methanogenic archaea as reported previously (Chan et al. 2005). In addition, the residual rates of CH_4 production upon inhibition with CH_3F increased with sediment depth, also indicating increasing f_{H_2}. Note that, as above, the calculation does not include further fractionation of the fermentatively produced acetate by acetoclastic methanogenesis. Interestingly, values of f_{H_2} were similar (45–55%) in the upper layers of the sediment core (Fig. 10) and in one grab sample (Fig. 8A), but larger than in two other grab samples at the end of incubation (16–32%; Fig. 8C), indicating that acetoclastic methanogenesis was relatively more important in the latter. We believe that this difference is real, since acetate concentrations were also higher in the latter (Fig. 8B) than in the former (below detection) samples. We assume that the difference was either due to the particular situation at the time of sampling (e.g., a sedimentation event) or to disturbance during grab sampling that stimulated acetate turnover. The latter assumption is reasonable, since radiotracer experiments have indicated that acetate turnover is compartmentalized in Lake Dagow sediment (Glissmann et al. 2004).

Another interesting feature is that f_{H_2} in the sediment cores was relatively high, with values of 40% to 67%. These values were all clearly above a value of $f_{H_2} = 33\%$, which is theoretically expected when organic matter, such as polysaccharides, is degraded methanogenically (Conrad 1999). If we come back to the possibility that δ_{ma} might be better represented by $\delta_{\text{ac-methyl-control}}$ than by $\delta_{\text{ac-methyl-}CH_3F}$, i.e., be more negative by about 5‰, then values of f_{H_2} would be lower by about 5%. The f_{H_2} in the upper sediment layers is then calculated to be on the order of 35–40% rather than 40–45%. This value would be much closer to the theoretically expected value. However, the deeper sediment layers still would exhibit a relatively high f_{H_2}. The occurrence of such high f_{H_2} is not without precedent (*see* review by Conrad 1999), but the reasons are unknown. There are several possible explanations. (1) The system is not really in steady state, so that fermentatively produced acetate and H_2 are methanogenically consumed at a different pace, e.g., acetate accumulating temporarily. This explanation can be excluded since acetate was not detectable. (2) Acetate is not only consumed by acetoclastic methanogens, but also by syntrophic acetate-consuming microbial consortia. In this process, acetate is first converted fermentatively to CO_2 and H_2, which is then converted to CH_4 by hydrogenotrophic methanogenesis. Evidence for this process has been found in sediment of Lake Kinneret (Nüsslein et al. 2001, 2003). However, this explanation is unlikely for Lake Dagow sediment, which contains a large population of acetoclastic *Methanosaeta* species (Chan et al. 2005; Penning et al. 2006*a*). (3) Small amounts of H_2 are produced in the sediment in addition to those produced during fermentation of polysaccharides, e.g., oxidation of reduced moieties within sediment organic material.

We think that the last one is the most likely explanation. In principle this means that organic matter would not be completely mineralized. Complete mineralization can be formulated as

$$C_6H_{12}O_6 + 2H_2O \rightarrow 2CH_3COOH + 2CO_2 + 4H_2 \quad (7)$$

$$4H_2 + CO_2 \rightarrow CH_4 + 2H_2O \quad (8)$$

$$2CH_3COOH \rightarrow 2CH_4 + 2CO_2 \quad (9)$$

$$\text{Net}: C_6H_{12}O_6 \rightarrow 3CH_4 + 3CO_2 \quad (10)$$

As is easily seen, complete mineralization under methanogenic conditions gives equal amounts of CO_2 and CH_4, the latter being formed from 1/3 hydrogenotrophic and 2/3 acetoclastic methanogenesis. Incomplete mineralization, on the other hand, may preferentially produce CH_4 rather than CO_2 and produce the CH_4 from H_2 plus CO_2 exclusively, e.g.,

$$C_6H_{12}O_6 + 2H_2O \rightarrow C_5H_{10}O_6 + CO_2 + 4H_2 \quad (11)$$

$$4H_2 + CO_2 \rightarrow CH_4 + 2H_2O \quad (12)$$

$$\text{Net}: C_6H_{12}O_6 \rightarrow C_5H_{10}O_6 + CH_4 \quad (13)$$

It is quite likely that part of the organic matter is incompletely oxidized even over very long time periods. In Lake Dagow sediment, 20-cm depth corresponds to an age of about 80 yr (Casper 1995). Organic matter in sediments can be much older than that (Mollenhauer and Eglinton 2007). The preferential conversion of organic carbon to CH_4 would eventually result in increase of the oxidation state of the residual organic matter. Such a change in oxidation state is difficult to assess, but is quite likely to occur. For example, the electron transfer capacity of organic matter in peat bogs was found to increase with depth (Heitmann et al. 2007). Another indication for change in oxidation state of organic matter is the relative increase of the rates of CH_4 production compared with the rates of CO_2 production, as seen from Eq. 13 compared with Eq. 12. This difference has been used for modeling electron balance in anoxic soils (Yao and Conrad 2000) but does not work in Lake Dagow sediment where CO_2 flux is affected by the high carbonate content of the sediment.

Our experiments showed that isotope fractionation factors can be quite different in Lake Dagow sediment when using different samples. Differences were even more pronounced when comparing the results of Lake Dagow sediment to those previously obtained in methanogenic rice field soil (Penning and Conrad 2007). Such differences are only revealed when isotope fractionation is studied in sufficient detail, which has hitherto rarely been done. Our studies in particular included measurement of $\delta^{13}C$ in acetate and acetate-methyl, but also included the explicit determination of α_{CO_2/CH_4} by inhibition of acetoclastic methanogenesis. With these data it was possible to quantify the hydrogenotrophic and acetoclastic paths of CH_4 production from stable isotopic signatures. We have demonstrated this application for a vertical sediment profile and found that hydrogenotrophic methanogenesis was relatively important in Lake Dagow sediment, especially in deeper sediment layers, indicating that organic matter was not completely degraded, leaving a relatively oxidized recalcitrant fraction. Our data also showed an unusual isotope fractionation during turnover of acetate, indicating that acetate production during fermentation of organic matter can exhibit different isotope fractionation factors and/or that acetoclastic methanogenesis can exhibit a nonnormal isotope fractionation ($\alpha < 1$). Finally, our data showed a rather large variation in the values of

α_{CO_2/CH_4}, covering a range from 1.043 to 1.097, which may be due to regulation by the in situ energetic conditions for hydrogenotrophic methanogenesis (Penning et al. 2005). This high potential variability demands the explicit determination of α_{CO_2/CH_4} for determining the relative contribution of H_2 plus CO_2 vs. acetate to total CH_4 production.

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